

Desigualdad triangular inversa para normas

Proposición 1 (Desigualdad triangular inversa). *Sea $(X, \|\cdot\|)$ un espacio normado*

$$\implies \forall x, y \in X : |||x| - |y|| \leq \|x - y\|.$$

Demostración: Por la métrica inducida/??, $\forall x, y \in X : d(x, y) := \|x - y\|$ define una métrica en X , luego por la desigualdad triangular inversa para métricas se cumple que

$$\forall x, y \in X : |d(x, 0) - d(y, 0)| \leq d(x, y) \implies |||x| - |y|| \leq \|x - y\|.$$

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Referenciado en

- prop-norma-continua

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